

Time-Resolved Spectral Gap Spectroscopy in a Quantum Simulator of Fermionic Superfluidity inside an Optical Cavity

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(Received 28 August 2024; revised 20 December 2024; accepted 3 April 2025; published 6 May 2025)

We use an ensemble of laser-cooled strontium atoms in a high-finesse cavity to cleanly emulate the technique of rf spectroscopy employed in studies of BEC-BCS physics in fermionic superfluids of degenerate cold gases. Here, we leverage the multilevel internal structure of the atoms to study the physics of Cooper pair breaking in this system. In doing so, we observe and distinguish the properties of two distinct many-body gaps, the BCS pairing gap and the spectral gap, using nondestructive readout techniques. The latter is found to depend on the populations of the internal atomic states, reflecting the chemical potential dependence predicted in fermionic superfluids. This work opens the path for more fully exploiting the rich internal structure of atoms in cavity QED emulators to study both analogous systems and also more exotic states yet to be realized.

DOI: [10.1103/PhysRevLett.134.183404](https://doi.org/10.1103/PhysRevLett.134.183404)

Ultracold atomic systems have emerged as a promising platform for simulating theories of quantum many-body physics, benefiting from the capability to engineer clean and tunable interactions of many forms [1–6]. An additional advantage lies in the complex internal structure of cold atoms, which have multiple ground and excited states and can lie beyond the paradigm of a traditional qubit. Such multilevel systems open up an even broader class of models and physical systems for study [5,7–17].

One iconic example is the development of ultracold Fermi gases using neutral atoms. Such experiments have enabled groundbreaking studies of the phenomenon of fermionic superfluidity, including a first realization of the BCS-BEC crossover [18–21], and have provided insight into a broad range of many-body systems [22,23]. To probe the superfluid pairing gap, early experiments relied on a “radio-frequency (rf) spectroscopy” technique, which involved weakly driving a mixture of two spin states along an rf transition to a (nominally noninteracting) third state [24–26]. This additional degree of freedom allowed researchers to observe an energy shift related to breaking Cooper pairs [27]. While this technique has been used for many purposes [28], undesired effects, such as competing pair-breaking processes and unwanted interactions between the third state and the other two states, complicated efforts to analyze the pairing gap [29–31]. Additionally, the

predicted shift, sometimes called the spectral gap Δ_{SG} , [32], depends nontrivially on the chemical potentials of the component spin states and is not equivalent to the pairing gap Δ_{BCS} [21,27].

Inspired by this context, here we extend earlier work studying BCS superconductivity in a cavity QED platform [33–35] by leveraging multiple internal levels of ^{88}Sr atoms. We start with implementing the so-called Anderson pseudospin mapping [36], mapping the presence and absence of a Cooper pair onto a long-lived electronic transition. The atoms, which are collectively coupled to a detuned optical cavity, experience an effective all-to-all interaction [37] which emulates the attractive and collective interaction between electrons in different momentum states in a superconductor. With this setup, we have previously studied the BCS Hamiltonian and observed dynamical phases in the pairing gap Δ_{BCS} [34]. By coupling to a third atomic state, we can mimic the setup of rf spectroscopy experiments and measure Δ_{SG} . Beyond the analogy to superfluidity, this system is also closely related to donor-acceptor models of energy transfer in biological systems, where the internal atomic states map onto donor and acceptor sites [38–40].

We construct an appropriate three-level system using an applied magnetic field to couple excited Zeeman states. By tuning the magnetic field strength, we explore this coupling both in a gapped regime and in a strong-coupling regime featuring large population transfer between states. Moreover, by varying the atomic inversion along the initial

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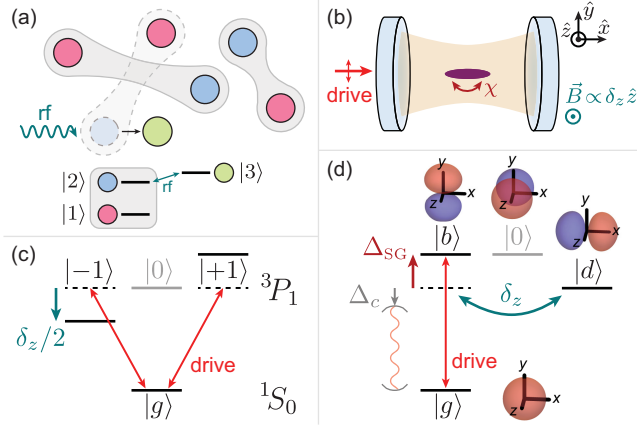


FIG. 1. Experimental setup. (a) The technique of “rf spectroscopy” used in ultracold fermion experiments measures a frequency shift, associated with breaking a Cooper pair, along an auxiliary transition. (b) In our system, ^{88}Sr atoms experience an effective infinite-range spin-exchange interaction through detuned collective coupling to an optical cavity, oriented along $\hat{x}$. We apply a tunable magnetic field along $\hat{z}$ and address the atoms with a $\hat{y}$ -polarized laser drive. (c) We explore a three-level system consisting of $|g\rangle = |^1S_0\rangle$ and $|\pm 1\rangle = |^3P_1, m_j = \pm 1\rangle$. We define δ_z as the splitting between $|\pm 1\rangle$ states. (d) The drive and cavity couple to the effective two-level system $|g\rangle \leftrightarrow |b\rangle$. In this basis, δ_z couples $|b\rangle$ to a third state $|d\rangle$, analogous to the level structure used in rf spectroscopy experiments. The cavity is detuned from the atomic transition by $\Delta_c = \omega_{c0} - \omega_a$; this generates a collective shift Δ_{SG} of $|b\rangle$, representing the spectral gap. Toy models of the atomic orbitals are shown to highlight the different polarizations required to address the excited states.

two-level system, we change both the number of particles participating in the pairing as well the pairing strength, allowing us to explore the distinction between Δ_{SG} and Δ_{BCS} . We accomplish all this through the use of two real-time probes, including a novel nondestructive, large dynamic range measurement of the cavity resonance frequency.

For these experiments, we cool and trap $N = 10^5$ – 10^6 ^{88}Sr atoms into a high-finesse optical cavity in the strong collective coupling regime, described in previous work [41,42]. The cavity’s unshifted resonance frequency ω_{c0} is red detuned from the bare $^3P_1 - ^1S_0$ transition frequency ω_a by $\Delta_c/2\pi := (\omega_{c0} - \omega_a)/2\pi = -51$ MHz. The relevant states include a single ground state $|g\rangle := |^1S_0\rangle$ and two long-lived ($\gamma/2\pi = 7.5$ kHz) excited Zeeman states $|\pm 1\rangle := |^3P_1, m_j = \pm 1\rangle$, with a quantization axis along $\hat{z}$ using the coordinates in Fig. 1(b). A $\hat{y}$ -polarized laser drive excites the “bright” superposition $|b\rangle_k := (1/\sqrt{2})(|+1\rangle_k + |-1\rangle_k)$ for each atom k , which also couples to the $\hat{y}$ -polarized cavity mode. The orthogonal “dark” state $|d\rangle_k := (1/\sqrt{2})(|+1\rangle_k - |-1\rangle_k)$ has a spatial distribution aligned with the cavity axis ($\hat{x}$), as shown in Fig. 1(d), and thus does not radiate into the cavity. In this

natural basis for our system, we define collective dipole operators $\hat{J}^+ = (\hat{J}^-)^\dagger := \sum_{k=1}^N |b\rangle\langle g|_k$, measuring dipole projections along $\hat{y}$, and the atomic inversion $\hat{J}^z = \sum_{k=1}^N \hat{J}_k^z$ where $\hat{J}_k^z := \frac{1}{2}(|b\rangle\langle b|_k - |g\rangle\langle g|_k)$. We also apply a tunable, uniform magnetic field $\vec{B} \parallel \hat{z}$, represented using the collective angular momentum operator $\hat{L}^z := \sum_{k=1}^N \frac{1}{2}(|+1\rangle\langle +1|_k - |-1\rangle\langle -1|_k) = \sum_{k=1}^N \frac{1}{2}(|b\rangle\langle d|_k + |d\rangle\langle b|_k)$ which describes linear Zeeman shifts.

Using the above definitions, up to atom-light coupling inhomogeneities neglected here for simplicity [43], we engineer an effective atomic Hamiltonian of the form

$$\hat{H}_a/\hbar = -\chi\hat{J}^+\hat{J}^- + \delta_z\hat{L}^z + \sum_{k=1}^N \epsilon_k|g\rangle\langle g|_k. \quad (1)$$

The first term describes an all-to-all cavity-mediated spin-exchange interaction with strength $\chi = g^2/\Delta_c$, where $2g/2\pi = 15.4$ kHz is the rms atom-cavity coupling as a single-photon Rabi frequency. Since this interaction is collective, it scales with the atom number with characteristic strength χN . δ_z is the tunable splitting between $|\pm 1\rangle$ Zeeman states; in the bright and dark basis, it creates an effective torque that rotates the collective dipole moment and couples $|b\rangle$ and $|d\rangle$. Finally, ϵ_k describes the differential light shift between the ground state $|g\rangle$ and the excited states due to the intracavity trapping light at 813 nm for atom k . This shift varies between the atoms due to a finite motional distribution [34], which induces dephasing between the ground and excited states.

The above Hamiltonian has a direct analogy to rf spectroscopy experiments using ultracold fermions in the BCS limit. Following an Anderson pseudospin mapping [36], we map the $|b\rangle$ and $|g\rangle$ states of atom k onto the presence and absence of a Cooper pair of fermions with different internal states $[|1\rangle$ and $|2\rangle$ in Fig. 1(a)] and momenta $\pm \mathbf{k}$. The spin-exchange term in Eq. (1) then represents the BCS interaction between fermions, and the ϵ_k term describes the electrons’ kinetic energy. Additionally, the (complex) BCS pairing gap Δ_{BCS} corresponds to $\chi\langle\hat{J}^-\rangle$; as previously shown, this quantity also sets the gap protection against single-particle dephasing in the case where $\langle\hat{J}^z\rangle = 0$ [34]. We can go further by mapping an atom in state $|d\rangle$ onto a “broken” Cooper pair consisting of one fermion in $|1\rangle$ and the other transferred to a third state $|3\rangle$. The Zeeman term coupling $|b\rangle$ to $|d\rangle$ then represents a drive which breaks up Cooper pairs [43]. Unlike the drive in rf spectroscopy experiments, this coupling operates at dc; we tune the coupling strength, rather than an rf drive frequency, in order to probe features of the spectral gap Δ_{SG} .

To study this model, we initialize the atoms with a collective drive angle $\theta_0 = \pi/2$ (defined as a $\pi/2$ pulse for maximally coupled atoms) along the $|g\rangle$ to $|b\rangle$ transition

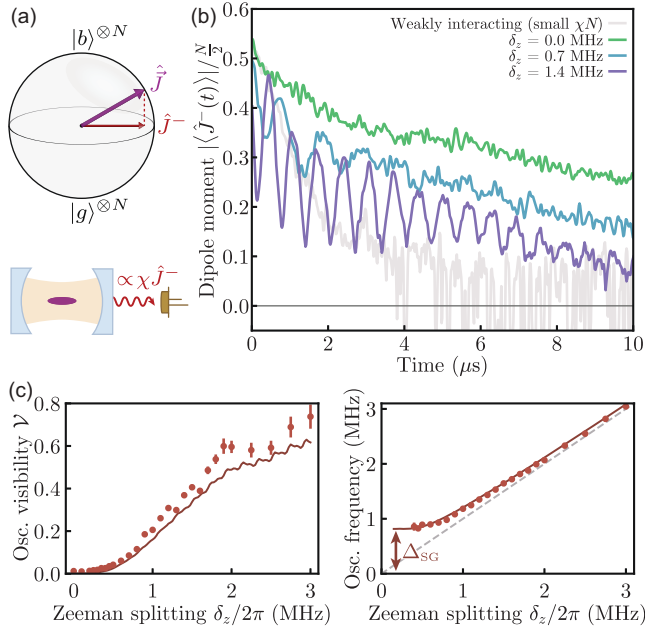


FIG. 2. Oscillations in the pairing gap Δ_{BCS} from excited state coupling. (a) The atoms emit light weakly into the cavity at a rate proportional to $\Delta_{\text{BCS}} = \chi \langle \hat{J}^- \rangle$, which we detect to infer dynamics. (b) Time traces of $|\langle \hat{J}^- \rangle|$ with $\chi N/2\pi = 1$ MHz. The Zeeman splitting δ_z induces rotations in the collective dipole moment, which are suppressed for small δ_z . The gray trace represents single-particle dynamics (with $\delta_z/2\pi = 0$ MHz), using a smaller N to weaken the collective interactions. (c) The average oscillation visibility (left) and frequency (right) from $t = 0.5$ μs to $t = 5$ μs for different δ_z . We also plot numerical simulations (solid curves) and the single-particle response $\omega_{\text{osc}} = \delta_z$ (gray dashed line). The dynamics resemble Rabi oscillation between excited states, with the Rabi frequency set by δ_z and the detuning generated by collective interactions which we identify as the “spectral gap” Δ_{SG} . To measure Δ_{SG} , we extrapolate the measured ω_{osc} to $\delta_z/2\pi \rightarrow 0$ MHz. All error bars in the paper represent $\pm 1\sigma$ deviations over a bootstrap resampling on experimental shots ($n_{\text{boot}} = 100$).

using a < 250 ns laser drive pulse. At time $t = 0$, we turn off the drive. As the atoms evolve under Eq. (1), they weakly emit $\hat{y}$ -polarized light out of the far-detuned cavity at a rate $\Gamma |\langle \hat{J}^- \rangle|^2$, where $\Gamma = |(\kappa/\Delta_c)\chi| \approx (|\chi|/300)$ sets the scale for collective emission [43]. As outlined in earlier work [34], this allows us to measure $\langle \hat{J}^- \rangle$ in real time by measuring this light as it leaks out of the cavity using a heterodyne detector [see Figs. 2(a) and 2(b)], with only a small fraction of atoms emitting light since $\Gamma \ll |\chi|$.

We first explore the competition between spin-exchange interactions and excited state coupling in Fig. 2(b) by comparing the time dynamics of the BCS pairing gap $|\Delta_{\text{BCS}}| = |\chi \langle \hat{J}^- \rangle|$ for different coupling strengths δ_z . Here, we scan $\delta_z/2\pi$ from 0 MHz to 5 MHz between shots while holding a fixed characteristic interaction strength $\chi N/2\pi = 1.0$ MHz. A background inhomogeneity set by $\{\epsilon_k\}$ with standard deviation $\epsilon/2\pi = 150$ kHz also remains

fixed; in the absence of interactions, this sets a dephasing time of 1 μs as shown by the gray trace. Consistent with previous results [34], we observe that sufficiently large interactions protect against dephasing at a scale set by χN , drastically enhancing the coherence time. The coupling induced by δ_z rotates the collective dipole moment, inducing oscillations in $|\langle \hat{J}^- \rangle|$. For small δ_z , weak oscillations open up at short times but decay after a few microseconds. In contrast, large δ_z traces feature large-amplitude oscillations with long lifetimes. The fact that $|\langle \hat{J}^- \rangle|$ remains large in this limit at long times shows that the interactions continue to protect against dephasing from the single-particle inhomogeneity $\{\epsilon_k\}$, even with a large δ_z .

Figure 2(c) further studies these two regimes by analyzing the oscillations in an interval from $t = 0.5$ μs to $t = 5$ μs . When $\delta_z/2\pi \gtrsim 2$ MHz, the oscillation visibility (defined as $\mathcal{V} = [|\langle \hat{J}^- \rangle|_{\text{max}} - |\langle \hat{J}^- \rangle|_{\text{min}}] / [|\langle \hat{J}^- \rangle|_{\text{max}} + |\langle \hat{J}^- \rangle|_{\text{min}}]$) is consistently large, and the oscillation frequency ω_{osc} approaches δ_z . For small δ_z , a different picture emerges: the visibility is suppressed and approaches 0, and ω_{osc} plateaus to a constant frequency. The experimental data agree with numerical simulations (solid curves) except for a small absolute scale factor in the oscillation visibility. This is likely related to additional dephasing mechanisms in our system (such as residual motional effects), which are unaccounted for in numerical simulations but would reduce the measured $|\langle \hat{J}^- \rangle|$ [34]. The oscillation frequency at small δ_z can be directly connected to the spectral gap Δ_{SG} observed in degenerate Fermi gas experiments when coupling to a third state. This is because our measurement of oscillations induced by a quenched DC coupling provides information equivalent to mapping out an rf spectroscopic peak.

An intuitive picture of the oscillations can be gained by imagining the atoms as electric dipoles. Atoms with coherence along the $|g\rangle$ to $|b\rangle$ transition exhibit a dipole moment proportional to $|\langle \hat{J}^- \rangle|$ which oscillates along $\hat{y}$ at optical frequencies. Applying a magnetic field along $\hat{z}$ creates an effective torque on the dipoles, causing them to rotate in the xy plane at a frequency set by the Zeeman splitting δ_z . Therefore, the population in $|b\rangle$ should periodically transfer between $|b\rangle$ (aligned with $\hat{y}$) and $|d\rangle$ (aligned with $\hat{x}$) while maintaining coherence with the ground state $|g\rangle$. This effectively causes the collective dipole moment $|\langle \hat{J}^- \rangle|$ along $\hat{y}$ to oscillate in time.

However, sufficiently strong spin-exchange interactions of the form $\hat{J}^+ \hat{J}^-$ disrupt this process. This term essentially shifts the excited state $|b\rangle$ by a frequency Δ_{SG} relative to $|d\rangle$, which does not radiate into the cavity. Quantitatively, $\chi \hat{J}^+ \hat{J}^- = \chi [\hat{J}^2 - (\hat{J}^z)^2 + \hat{J}^z]$ has eigenstates $|J, J^z\rangle$ and eigenvalues $E_{J, J^z} = \chi [J(J+1) - J^z(J^z-1)]$. For fully collective states, i.e., where $J = [(N_b + N_g)/2]$ and $J^z = [(N_b - N_g)/2]$ with $N_{b,g}$ denoting the populations in $|b\rangle$

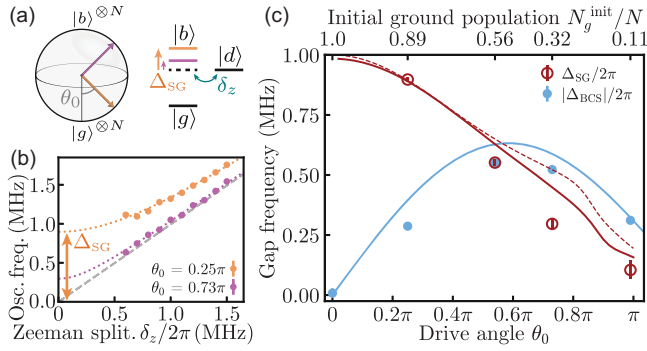


FIG. 3. Probing the spectral gap Δ_{SG} . (a) As the drive angle θ_0 increases, N_g decreases, which should reduce the spectral gap Δ_{SG} . (b) We repeat the experiment in Fig. 2(c) for different drive angles θ_0 . For each θ_0 , we fit the results to the form $\omega_{osc} = \sqrt{\Delta_{SG}^2 + \delta_z^2}$ (dotted lines) and extract Δ_{SG} . (c) Measured values of Δ_{SG} [as outlined in (b)] and $|\Delta_{BCS}|$ (at short times $t = 0$ μ s to $t = 0.5$ μ s) for different θ_0 , or alternatively equivalently the $t = 0$ ground state population N_g^{init} (circles). The two gaps overlap at $\theta_0 = 0.54\pi$. We compare against numerical simulations of the two gaps (solid curves) and χN_g^{eff} (dashed curve), where N_g^{eff} is the ground state population weighted by cavity coupling and averaged over the measured time interval $t = 0.5$ μ s to $t = 5$ μ s.

and $|g\rangle$ respectively, $E_{J,z} \approx (\chi N_g) N_b$ in the large- N limit. Therefore, transferring one atom from $|b\rangle$ to $|d\rangle$ imposes an energy penalty equal to $\hbar \Delta_{SG} = \hbar \chi N_g$. Note that Δ_{SG} decreases with increasing atomic inversion, mirroring the chemical potential dependence in degenerate Fermi gases.

To explore this dependence in our system, we repeat the above experiment with different initial drive angles θ_0 (for maximally coupled atoms) along the $|g\rangle$ to $|b\rangle$ transition. As Fig. 3(a) illustrates, increasing θ_0 decreases the initialized ground state population N_g^{init} , which should reduce the splitting between the excited states and therefore Δ_{SG} . We see this trend in Fig. 3(b): when $\theta_0 = 0.73\pi$, the oscillation frequency resembles the expected single-particle response, whereas the $\theta_0 = 0.25\pi$ data exhibit large frequency deviations which indicate a sizable spectral gap Δ_{SG} . We determine Δ_{SG} by fitting the data to the form $\omega_{osc} = \sqrt{\Delta_{SG}^2 + \delta_z^2}$, the predicted response for homogeneous coupling [43], over the domain $\delta_z/2\pi \geq 0.6$ MHz where the oscillations are most prominent (dotted lines).

In Fig. 3(c), we observe that Δ_{SG} and $|\Delta_{BCS}|$ scale differently with the drive angle θ_0 : Δ_{SG} monotonically decreases, and $|\Delta_{BCS}|$ starts at 0 and peaks around $\theta_0 \approx 0.6\pi$. These qualitative behaviors are consistent with the expected relations $\Delta_{SG} \propto N_g$ and $\Delta_{BCS} \propto \langle \hat{J}^- \rangle$. Numerical simulations calculated using a similar analysis (solid curves) agree qualitatively with experimental values but overestimate some points. We attribute this to the additional dephasing mechanism described earlier, which would weaken the interaction strength and therefore reduce the size of the gaps.

We also directly compare Δ_{SG} with numerical simulations of χN_g for different θ_0 [dashed curve in Fig. 3(c)]. Complications arise in this calculation from inhomogeneous atom-light coupling, since the effective N_g seen by the cavity varies in time (unlike in homogeneously coupled systems where it does not vary on timescales faster than the decay rate). To account for this, we weight atoms by cavity coupling and average the result over the measurement interval ($t = 0.5$ μ s to $t = 5$ μ s) to obtain an effective shift χN_g^{eff} (see Supplemental Material [43]). This curve qualitatively resembles the measured values of Δ_{SG} , demonstrating the expected scaling.

We also directly probe internal state population dynamics using a nondestructive measurement of the atomic inversion $\langle \hat{J}^z \rangle$. In addition to the atomic Hamiltonian in Eq. (1), the total effective Hamiltonian includes an atom-cavity contribution [45]:

$$\hat{H}_{ac}/\hbar = (\Delta_c - 2\chi \hat{J}^z) \hat{a}^\dagger \hat{a}, \quad (2)$$

where $\hat{a}$ is the annihilation operator for the $\hat{y}$ -polarized cavity mode. The cavity resonance frequency $\omega_c(t)$ then varies in response to $\langle \hat{J}^z \rangle$, allowing us to infer the dynamics of $\langle \hat{J}^z \rangle$ by measuring $\omega_c(t)$. Our probe scheme, explained fully in the Supplemental Material [43], involves applying 100 ns laser pulses with center frequency ω_{c0} every 5 μ s, similar to work performed in a resonant atom-cavity system [46]. As illustrated in Fig. 4(a), the short pulse length induces Fourier broadening, allowing excitation of the cavity mode even if $\omega_c(t)$ deviates by several MHz. This feature gives our probe a large dynamic range for $\omega_c(t)$. After each pulse, the cavity freely rings down, and the leakage “ringdown” light exhibits a frequency that adiabatically follows $\omega_c(t)$ [Fig. 4(b)], which we measure to resolve fast changes in $\omega_c(t)$.

Figure 4(c) shows time dynamics of $\langle \hat{J}^z \rangle$ extracted from our probe. Before the sequence begins, $\langle \hat{J}^z \rangle = -N/2$. An initialization pulse at $t = 0$ with drive angle $\theta_0 = 0.5\pi$ rapidly increases $\langle \hat{J}^z \rangle$. Afterward, we observe oscillations in $\langle \hat{J}^z \rangle$ in some datasets. For smaller δ_z , these oscillations are hard to resolve above the probe noise floor, whereas larger δ_z sets display more prominent oscillations. Assuming N_g does not change significantly, oscillations in $\langle \hat{J}^z \rangle / (N/2) = (N_b - N_g)/N$ reflect transfer between $|b\rangle$ and $|d\rangle$. We characterize this process by plotting the oscillation visibility for different δ_z relative to a baseline at $\langle \hat{J}^z \rangle = -N/2$ [orange points in Fig. 4(d)]. Below $\delta_z/2\pi = 1$ MHz, the lack of visible oscillations above the noise floor indicates poor population transfer. Above this point, however, the visibility sharply rises and plateaus above 2 MHz, suggestive of large oscillations between $|b\rangle$ and $|d\rangle$. The experimental data agree well with numerical simulations (solid lines).

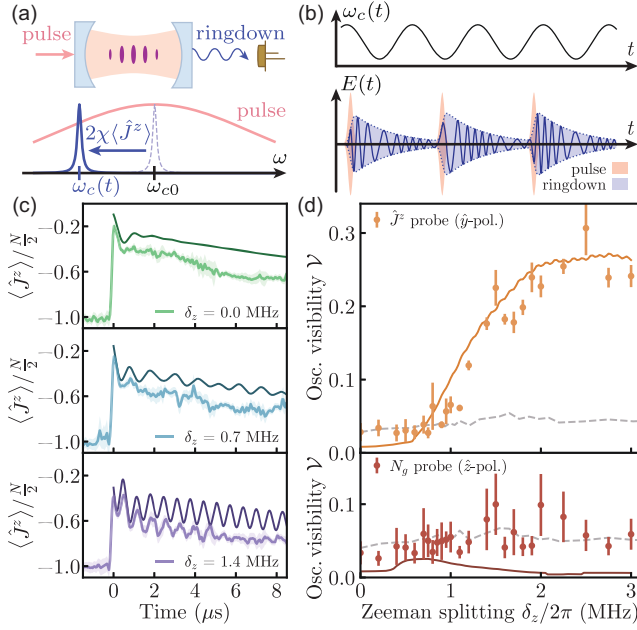


FIG. 4. Directly measuring excited state population transfer. (a) We probe the cavity resonance by sending light pulses at the cavity and detecting the transmitted light (“ringdown”). The pulses resonantly excite the cavity over a large range of $\omega_c(t) = \omega_{c0} - 2\chi\langle J^z \rangle$. (b) The intracavity field adiabatically follows $\omega_c(t)$, which can be measured in ringdown to infer $\langle J^z \rangle$. (c) Time traces of $\langle J^z \rangle$ (colored curves) with $\pm 1\sigma$ bounds for different δ_z , compared with numerical simulations (thinner dark curves). $\langle J^z \rangle$ is estimated in 50 ns time bins and then smoothed with a Gaussian filter ($\tau_{\text{smooth}} = 50$ ns). (d) Top: $\langle J^z \rangle$ oscillation visibility (circles) from $t = 0.5$ μ s to $t = 5$ μ s, relative to a baseline at $\langle J^z \rangle = -N/2$, alongside numerical simulations (solid curve) and a noise floor (dashed line) inferred from data at $\delta_z/2\pi = 0$ MHz. The noise floor is not flat because calculating visibility involves normalizing each data point separately. Bottom: a similar analysis for an orthogonal $\hat{z}$ -polarized probe, which is sensitive to N_g instead of $\langle J^z \rangle$. The noise floor is consistent with the top plot to aid in a visual comparison of the two probes.

To confirm that the oscillations in $\langle J^z \rangle$ arise from dynamics in N_b and not N_g , we also use a $\hat{z}$ -polarized cavity probe [red points in Fig. 4(d)]. In analogy with the $\hat{y}$ -polarized probe, this probe measures oscillations of $(N_0 - N_g)/N$, where N_0 is the population in $|^3P_1, m_j = 0\rangle$. Since we nominally do not excite this state, we assume N_0 is small, such that this probe solely estimates changes in N_g . We do not measure an oscillation visibility significantly above the noise floor, so we conclude that the observed oscillations in $\langle J^z \rangle$ primarily represent population transfer between $|b\rangle$ and $|d\rangle$.

In this work, we have clarified the difference between two distinct many-body gaps, Δ_{BCS} and Δ_{SG} . These results are enabled by utilizing the relative advantages of cavity QED for studying superfluid systems: our system has direct access to multiple relevant observables, including the

pairing gap Δ_{BCS} . From a technical standpoint, the system exhibits tunable interactions without requiring a Feshbach resonance that induces significant two- and three-body losses, and our stronger interaction strength allows us to neglect the effects of vacuum loss.

By starting with a pristine emulator of the BCS model with well-determined parameters, we show that we can add complexity in the form of controllable pair-breaking terms. In the future, we can add further complexity to explore features such as higher interaction orders of superfluidity [47]. This experiment also shows the potential promise in leveraging multiple internal states for exploring more complex physics, such as correlated hopping processes between nuclear sublevels in ^{87}Sr [13] or generation of multilevel entangled dark states [14–16].

Acknowledgments—We thank Junyu Lin, Maya Miklos, and Jun Ye for stimulating discussions and a careful reading of the manuscript. This material is based upon work supported by the U.S. Department of Energy, Office of Science, National Quantum Information Science Research Centers, Quantum Systems Accelerator. We acknowledge additional funding support from the Vannevar Bush Faculty Fellowship, the National Science Foundation under Grants No. PFC PHY-2317149 (Physics Frontier Center), No. OMA-2016244 (QLCI Q-SEnSE), and NIST. D.B. was supported by the Simons Collaboration on Ultra-Quantum Matter (UQM) which is funded by grants from the Simons Foundation (Grant No. 651440), and acknowledges support in part by Grant No. NSF PHY-2309135 to the Kavli Institute for Theoretical Physics (KITP). R. J. L.-S. acknowledges support by the Air Force Office of Scientific Research under Award No. FA9550-24-1-0106.

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